SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 2 – Concept Mapping (5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO1	**Assessment:**	Assessment 2: Concept Mapping
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO1 Statement:	***Apply the fundamental concepts, image formation models, and image processing techniques to analyze and solve real-world Computer Vision problems.***		
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Section / Batch:	CSE - Finalyear	**Date:**	05/10/26

Code of Conduct

I M. Neela Meghana 192311002 *(NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: M

QUESTIONS: 5

Total Marks: 50

Annexure A

Concept Mapping

Q1. An automobile company is developing an autonomous vehicle that uses cameras to detect vehicles, pedestrians, and traffic signs. Objects closer to the camera appear larger, while distant objects appear smaller.
Prepare a concept map applying the image formation and camera projection model to show how a three-dimensional road scene is converted into a two-dimensional image. Include the relationships between focal length, object distance, image distance, and projected image size.

Q2. A manufacturing company uses cameras to measure the dimensions of bottles moving on a conveyor belt. Engineers observe that bottles at different distances from the camera appear to have different sizes.
Prepare a concept map applying perspective projection to show how focal length, camera position, viewing angle, object distance, and image size are related. Use the concept map to represent how these factors influence the accuracy of bottle measurement.

Q3. A construction company uses a drone camera to capture images of buildings from different heights and viewing angles. The appearance and measured dimensions of the building change when the drone position is changed.
Prepare a concept map applying image formation and projection concepts to show the relationship between camera orientation, focal length, perspective projection, camera position, calibration, and the accuracy of building measurements.

Q4. An industrial company wants to select a suitable projection model for road monitoring and inspection of objects placed on a flat conveyor belt. The engineers are considering perspective and orthographic projection for these applications.
Prepare a concept map comparing perspective and orthographic projection based on depth, object size, distance, parallel lines, and image formation. Use the concept map to identify the appropriate projection model for each application.

Q5. A robotics company wants a robot to identify the position and shape of objects placed on a table using a camera. The camera captures only a two-dimensional representation of the three-dimensional objects.
Prepare a concept map applying the image formation model to show the relationship between a 3D object, camera parameters, projection geometry, focal length, camera orientation, and the resulting 2D image. Show how these concepts help determine the object's position and appearance.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: 5/10/26.	Date: _____	Date: _____

3D Road Scene (Real World)

- Consists of real-world coordinates (x, y, z).
- z = Object distance (depth from the camera lens)

captured by [Pin-hole projection]

Camera projection Model

- Maps continuous 3D world space points onto a continuous 2D
- Governed by ray geometry intersecting at a single optical center.

determines

governed by

Object Distance (z)

- Distance from object to lens center.
- Key variable in depth calculation.

Focal length (f)

- Distance from lens center to image plane sensor.

combines via Thin lens Eqⁿ $1/f = 1/z + 1/v$

Image Distance (v)

- Distance behind the lens where lights rays perfectly coverage.
- For distant road scenes ($z \gg f$), image distance coverages near $v \approx f$.

determines Magnification $M = -v/z \approx -f/z$

Projected Image Size (h')

- Formula : $h' = (f \times h) / z$ [h - true object height].

creates

Visual Phenomenon

2D Projected Road Image

- closer Objects : High mag → Large ; Distant : Low Mag → Small

Conveyer Belt Measurement setup

- System measures moving items using dynamic visual feeds.
- Goal: Maintain highly uniform, sub millimeter inspection accuracy.

guided by [Geometric shape]

Perspective Projection

- Projects 3D scene rays to a central point, creating inherent perspective scaling where object size varies dynamically based on real depth.

fixed by

Focal length (f)

- Length of lens
- Larger f narrows FOV but magnifies obj shape

fixed by

Camera Position

- Height & distance relative to the conveyer ~~tracks~~

fixed by

Object Distance (z)

- Physical gap from the lens to the inspected bottle.

defines [sensor view span]

Viewing Angle (FOV)

- Determined jointly by lens focal length (f) and sensor size.
- Wide angles compress edges; narrow fields minimize radial distortion.

projects to

[Resulting Output Size]

Projected Image Size

- Pixels allocated per bottle image width / height : scale = f/z
- Minor vertical bouncing on belt changes z, altering pixel size.

impacts

Bottle Measurement Accuracy

- Problem : Parallax and distance variations introduce sizing errors.
- Fixes : Standardize object distance (z) or utilize telecentric lenses

Drone-Based Building Inspection & Photogrammetry

Drone Inspection Platform

- Mobile aerial vehicle platform housing an imaging payload.
- Faces variations in position (x, y, z) and orientation angles.

↓ experiences

Camera Position & Orientation

- Position : Alters objects distance, shifting image scale.
- Orientation (Roll / Pitch / Yaw) : skews right angle into oblique views.

↓ dictates

Perspective Projection

- causes geometric foreshortening
- Converging parallel structural lines.

↓ fixes

Focal length (f)

- Hardware baseline parameter
- Determines zoom factor and global ground sampling distance

↓ To resolve Ambiguity

Camera Calibration

- calculates Extrinsics (R, T matrices) & Intrinsics (f, principal pt).
- Corrects for barrel / pincushion lenses radial distortion artifacts.

↓ ensures [High-Quality Mapping]

Building Accuracy

- Converts 2D pixel coordinates back to real-word metric dimensions.

4. Comparison of Projection Models [Perspective Vs Orthographic]

Projection Model Analysis

- Purpose : Determine mapping from 3D world space coordinates onto 2D sensor grids.

evaluates ↓

Perspective Projection

- Rays intersect at central projection origin

evaluates ↓

Orthographic Projection

- Rays project completely parallel to axis.

↓

Comparison Criteria

1. Depth Perception
   - Perspective : Retains distinct depth cues; distant items shrink.
   - Orthographic : Eliminates depth cues entirely; independent of z-coordinate.
2. Object Size & Distance
   - Perspective : Size scales inversly with distance (size $\propto 1/z$).
   - Orthographic : size remains perfectly constant regardless item distance.
3. Parallel Lines
   - Perspective : Converge to a vanishing point
   - Orthographic : Affine mapping variant assuming infinite camera d.

↓ optimum for

Application 1 : Road Monitor

- Wide fields of view with deep depth range.
- Models incoming traffic size.

↓

Application 2 : Conveyer Inspection

- Inspected flat parts at constant distance.
- Eliminates magnification shifts

Robotic TableTop Manipulation Object Localization.

3D object on a Table

- Possesses physical bounding dimensions, real shapes, and poses.
- Defined by it's 3D coordinates relative to the table workspace origin.

↓ Observed by [Optical Systems]

Projection Geometry

- Maps 3D space vectors into 2D spaces : $x = k * [R|T] * X$
- Replaces real 3D coordinates with a flattened pixel array

↓ governed by [Hardware structure]

Intrinsics

- Internal specs : Focal length (f), aspect ratios, & principal point

↓ governed by [Spatial setup]

Extrincis

- 3D rotation and translation angles relative to table boundary

↓ back-calculated [solving Inverse Problem]

Resulting 2D Image

- Flattened pixel canvas representing object borders and shadows.
- Object shape features skew based on the viewing angle.

↓ Produces Flat Media O/P.

Position & Appearence Inference

- Inverse Projection : Reconstruct 3D metrics from pixel inputs.
- Uses homograph constraints to extract real positions.